Block 5 – Session 1

Software-Defined & Cloud-Native Networking

Arquitectura de Xarxes

Universitat Pompeu Fabra

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Outline

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What If You Could Program Your Entire Network?

Traditional networks are configured device by device.
What if the network was as programmable as software?

Learning objectives:

1. Explain SDN architecture and why it matters
2. Understand NFV and how it replaces hardware appliances
3. Describe container networking and microservices communication

Traditional Network Management

- Each device (router, switch, firewall) configured **individually**
- Configuration via CLI, **device by device**
- Vendor-specific commands and interfaces
- Changes require **manual intervention** on every device

Imagine updating 1,000 switches one by one. That's traditional networking.

Scalability Challenges

- Data centers grow from hundreds to **hundreds of thousands** of devices
- Manual configuration does **not scale**
- Network changes are **slow**: days or weeks for approval + implementation
- **Human errors** increase with complexity
- Cloud-scale infrastructure demands **automation**

The Control Plane Problem

- In traditional networks, **every device** runs its own control plane
- Each router independently computes routes (OSPF, BGP from Block 3)
- No **centralized view** of the entire network
- Difficult to implement **network-wide policies**
- Limited **programmability**: cannot easily add new features

The control plane is **distributed** by design. What if we **centralized** it?

Discussion: The Cost of Manual Networks

Why have traditional networks survived so long despite their scalability problems?

- Hint: think about reliability, vendor relationships, and risk aversion
- What would it take for an organization to change?

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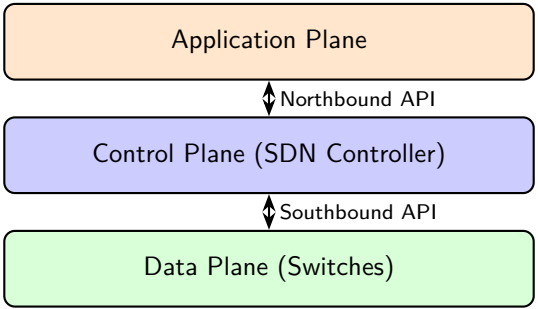
SDN – Core Idea

- **Separate** the control plane from the data plane
- **Centralize** network intelligence in a software controller
- **Program** network behavior through APIs



McKeown et al., "OpenFlow: Enabling Innovation in Campus Networks," *ACM SIGCOMM CCR*, 2008.

SDN Architecture – Three Planes

Data Plane (Infrastructure Layer):

- Physical/virtual switches that **forward packets**
- No longer make independent routing decisions
- Receive forwarding rules from the controller

Control Plane (Controller):

- Centralized software with a **global view** of the network
- Computes paths, installs forwarding rules on switches
- Single point of management for the entire network

Application Plane:

- Applications that define **network behavior** (firewall, load balancer, monitor)
- Communicate with the controller via northbound APIs

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SDN Controllers

- The controller is the **brain** of an SDN network
- Maintains a real-time **topology database** of the entire network
- Supports **high availability** (clustered deployment)

Examples:

- **OpenDaylight** (Linux Foundation, Java-based, widely adopted)
- **ONOS** (Open Network Operating System, carrier-grade, telecom-focused)
- **Floodlight** (open-source, lightweight, educational)
- Proprietary: **Cisco ACI**, **VMware NSX**

OpenDaylight Project, opendaylight.org; ONOS Project, onosproject.org.

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Southbound Interface – OpenFlow

- **OpenFlow** (2008, Stanford): the original SDN protocol
- Defines how the controller communicates with switches
- Controller installs **flow rules** in the switch's flow table:

Match	Action	Priority
dst IP = `10.0.1.0/24`	Forward to port 3	100
dst IP = `10.0.2.0/24`	Forward to port 5	100
src IP = `192.168.1.100`	Drop	200
* (any)	Send to controller	1

- If no rule matches → packet sent to controller for a decision

ONF, "Software-Defined Networking: The New Norm for Networks," ONF White Paper, 2012.

Northbound Interface – Representational State Transfer (REST) APIs

- Applications interact with the controller via **REST APIs**
- Example operations:
  - `GET /topology/links` → get network topology
  - `POST /flows` → add a flow rule
  - `GET /statistics/port` → query traffic statistics

Key benefit: the network becomes **programmable** like any other software system.

Any developer can write applications that control network behavior.

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SDN Benefits – Summary

- **Centralized management**: one controller, global network view
- **Programmability**: automate changes via APIs
- **Agility**: deploy new policies in seconds, not days
- **Vendor independence**: OpenFlow works across vendors
- **Innovation**: researchers and developers can experiment easily

SDN is the foundation for modern cloud networking: AWS, Azure, and GCP all use SDN internally.

Discussion: SDN Trade-offs

If the SDN controller fails,
what happens to the network?

- Hint: single point of failure vs distributed control
- How do real deployments address this? (clustering, failover)

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From Hardware Appliances to Software

- Traditional network functions run on **dedicated hardware**:
  - Firewall → physical appliance ($10K–$100K+)
  - Load balancer → physical appliance
  - Router → physical appliance
- Problems: **expensive**, inflexible, vendor lock-in, slow to deploy
- NFV idea: run these functions as **software on commodity servers**

ETSI, "Network Functions Virtualisation – An Introduction, Benefits, Enablers, Challenges & Call for Action," White Paper, 2012.

Virtual Network Functions (VNFs)

- A **VNF** = network function implemented as software
- Runs on VMs or containers, on standard x86 servers

Physical Appliance	**VNF Equivalent**
Hardware firewall	Virtual firewall (pfSense, iptables)
Hardware load balancer	Virtual LB (HAProxy, NGINX)
Hardware router	Virtual router (VyOS, FRRouting)
WAN optimizer	Virtual WAN optimizer
Intrusion Detection / Prevention (IDS/IPS)	Virtual IDS (Snort, Suricata)

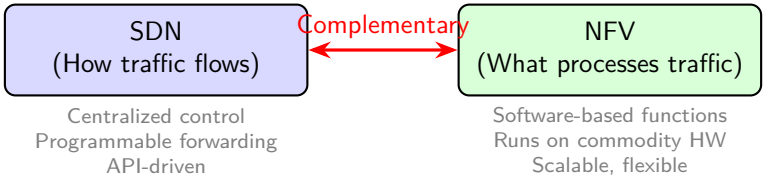
ETSI GS NFV 002, "NFV Architectural Framework," v1.2.1, 2014.

NFV + Cloud Integration

- Deploy VNFs on cloud infrastructure (IaaS) $\rightarrow$ **minutes** instead of months
- **Scale horizontally**: add more instances under load
- **Update easily**: software upgrades, no truck rolls
- **Cost reduction**: commodity servers vs specialized hardware

No hardware procurement, no shipping, no rack-and-stack.

SDN + NFV – Complementary Technologies



- SDN **steers** traffic to the right VNF (forwarding rules)
- NFV **processes** the traffic (filter, balance, encrypt, inspect)

Discussion: When Hardware Still Wins

Can you think of a scenario where a physical appliance is better than a VNF?

- Hint: think about latency, throughput, and specialized workloads
- What about hardware accelerators (FPGAs, SmartNICs)?

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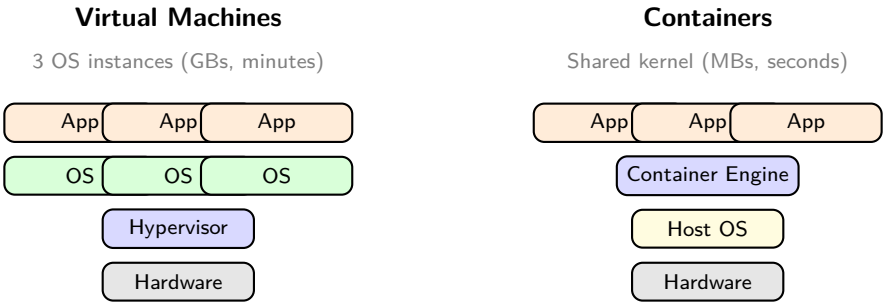
Containers – Deep Dive (Expanding Block 4 Overview)

- In Block 4 we compared VMs vs containers at a high level
- Now: **how containers actually work**
- Containers share the **host kernel** (unlike VMs with full guest OS)
- Isolation via Linux kernel features:
  - **Namespaces**: isolate process trees, network stack, filesystem
  - **cgroups**: limit CPU, memory, I/O per container
- **Docker**: most popular container platform
  - Packages app + dependencies into a **container image**
  - Image runs as a **container**: lightweight, portable, reproducible

Docker, Inc., "Docker Overview," docs.docker.com; OCI, "Open Container Initiative," opencontainers.org.

VM vs Container – Revisited



- VMs: **minutes** to start, **GBs** in size, 3 full OS running
- Containers: **milliseconds** to start, **MBs** in size, shared kernel
- In practice: containers often run **inside** VMs for extra isolation

Container Network Models

Bridge network (default):

- Containers on the same host connected via a **virtual bridge** (`docker0`)
- Each container gets a **veth pair** (virtual Ethernet interface)
- **NAT** for external traffic; containers share host's IP

Host network:

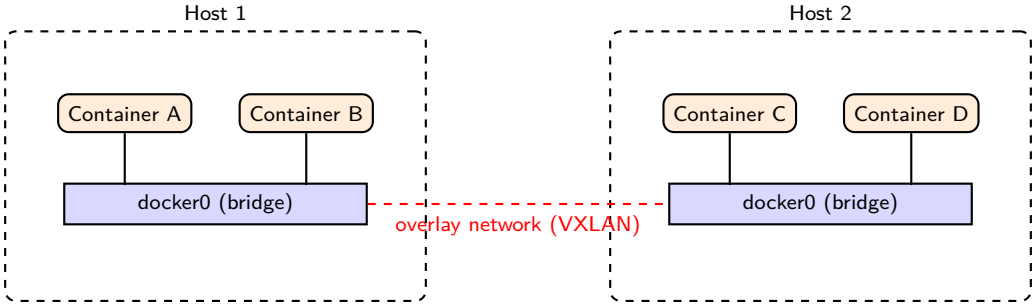
- Container uses the **host's network stack directly**
- No network isolation, but **no NAT overhead**
- Use for performance-critical applications

Overlay network:

- Containers on **different hosts** connected via VXLAN overlay
- Same overlay concept from Block 4, now applied to containers
- Enables **multi-host** container deployments

Docker, Inc., "Docker Networking Documentation," docs.docker.com/network.

Container Networking – Visual



- Within a host: containers use a **bridge** network
- Across hosts: an **overlay** connects the bridges (same VXLAN from B4)

Microservices – From Monolith to Distributed

- **Monolithic** app: one big process, one deployment, one codebase
- **Microservices**: app split into small, independent services
  - Each service runs in its **own container**
  - Services communicate over the **network** (HTTP/REST, gRPC Remote Procedure Call)

Network implications:

- Hundreds of containers talking to each other
- Need **service discovery**: "where is the payment service?"
- Need **load balancing**: distribute requests across replicas
- Need **observability**: trace requests across services

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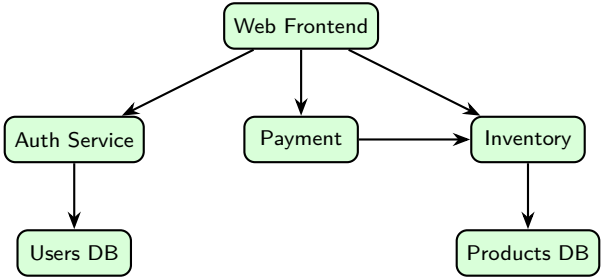
Microservices – Network Challenges



- Every arrow = **network call** → latency and failure risk
- If Inventory is down → Payment and Web are affected (**cascading failure**)
- Solutions: retries, timeouts, circuit breakers, service meshes

Discussion: Containers vs VMs

If containers are faster and lighter than VMs,
why do companies still use VMs?

- Hint: security isolation, legacy applications, compliance requirements
- In practice, containers often run inside VMs

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Infrastructure as Code (IaC)

- **IaC**: manage infrastructure through **code files**, not manual clicks
- Describe desired state in a configuration file → tools apply it automatically

Key benefits:

- **Reproducibility**: same config → same infrastructure, every time
- **Version control**: track changes in Git, review, rollback
- **Automation**: no manual steps → fewer human errors
- **Speed**: deploy entire environments in minutes

Morris, *Infrastructure as Code*, 2nd ed., O'Reilly, 2021.

IaC Example – Terraform (Conceptual)

```
# Define a VPC
resource "aws_vpc" "main" {
  cidr_block = "10.0.0.0/16"
}

# Define a subnet
resource "aws_subnet" "public" {
  vpc_id     = aws_vpc.main.id
  cidr_block = "10.0.1.0/24"
}

# Define a security group
resource "aws_security_group" "web" {
  vpc_id = aws_vpc.main.id
  ingress {
    from_port = 80
```

Discussion: IaC Risks

If infrastructure is defined as code,
what happens when someone pushes a bug?

- Hint: code review, staging environments, automated testing
- How is this similar to software development best practices?

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Serverless Computing (FaaS)

- **Serverless** (Function as a Service, FaaS):
  - Deploy individual **functions**, not entire servers
  - Triggered by **events** (HTTP request, file upload, timer)
  - Provider manages **all infrastructure**: zero server management
  - Pay only for **execution time** (measured in milliseconds)
- Examples: **AWS Lambda**, **Azure Functions**, **Google Cloud Functions**

Trade-offs: less control, potential vendor lock-in, cold start latency.

AWS, "AWS Lambda Documentation," docs.aws.amazon.com/lambda.

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Edge Computing

- **Edge computing**: process data **close to where it is generated**
- Instead of sending everything to a central cloud → process at the edge
- Benefits:
  - **Lower latency** (critical for IoT, autonomous vehicles, AR/VR, gaming)
  - **Less bandwidth** (filter and aggregate data locally)
  - **Better privacy** (data stays closer to the source)

Cloud and edge are **complementary**, not competing:

- Edge: real-time, latency-sensitive tasks
- Cloud: heavy computation, long-term storage, analytics

ETSI, "Multi-access Edge Computing (MEC)," etsi.org/technologies/multi-access-edge-computing.

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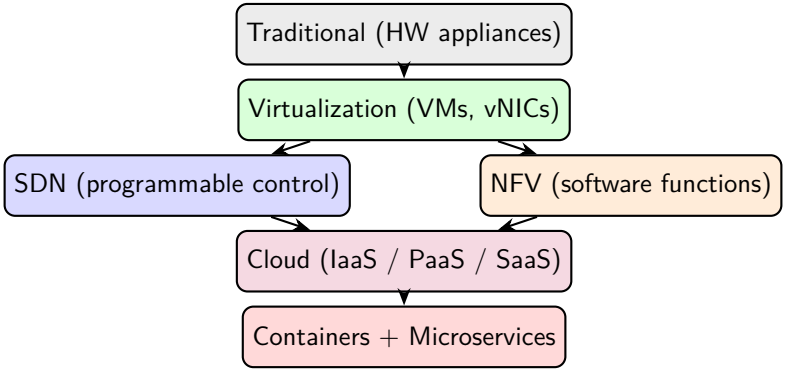
The Full Picture – Where Everything Fits



Each layer builds on the previous one. This is the **evolution of networking**.

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Key Takeaways

1. Traditional networks are **manual, rigid, and do not scale**
2. **SDN** separates control from data plane $\rightarrow$ centralized, programmable
3. **NFV** replaces hardware appliances with software $\rightarrow$ flexible, cheap
4. SDN + NFV are **complementary**: SDN steers, NFV processes
5. **Containers** are lightweight (shared kernel), networked via bridges/overlays
6. **Microservices** = many containers communicating over the network
7. **IaC** automates infrastructure $\rightarrow$ reproducible, fast, auditable

Discussion

A company runs 200 microservices in containers.
One Friday at 5 PM, a manual network change
breaks communication for 50 of them.
How could SDN and IaC have prevented this?

Hints: centralized control, automated testing, instant rollback, reproducibility.